PROTON+: Contamination-Robust Prototype Banks for Test-Time Online Out-of-Distribution Detection with Medical Vision-Language Models

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Abstract

Medical vision-language models are deployed as zero-shot classifiers and must flag inputs that lie outside the classes they were prompted with. Static detectors such as maximum concept matching (MCM) read only the softmax over the class prompts. An image from a new camera or a new population is classified confidently, so the softmax carries no trace of the shift. PROTON answers this with an online bank of class prototypes filled from confident test predictions, and fuses the distance to the nearest prototype with MCM. We reproduce PROTON from scratch on retinal photographs, chest radiographs and histopathology patches with three medical vision-language models, and find that its confidence gate admits exactly the confidently misclassified shifted inputs, so the prototypes absorb the shift and the detector can fall below MCM. PROTON+ keeps the online bank but anchors every class prototype to the dominant imaging-domain mode of the bank with a mean-shift step, admits only samples that are consistent with their class prototype, and fuses online z-scores instead of raw scores. With the hyperparameters of PROTON unchanged, PROTON+ improves over MCM on covariate and far shifts in all three modalities and closes most of the gap to an oracle bank that never sees a shifted sample. On covariate shift PROTON+ raises AUROC over MCM by 47.3 points on chest radiographs and by 16.4 points on ultra-widefield fundus photographs.

1 Introduction

Medical vision-language models (VLMs) such as `FLAIR`, `UniMed-CLIP` and `QuiltNet` classify an image from text prompts alone, without task-specific training [7, 12, 23]. A screening service prompted with "normal" and "diabetic retinopathy" also receives glaucoma cases, ultra-widefield photographs from a different camera, and images that are not retinas at all. Each of these inputs must be flagged for review instead of being assigned to one of the two classes. Out-of-distribution (OOD) detection assigns every input a score that is higher for inputs the classifier should not decide on, and a detector is judged by how well that score separates in-distribution (ID) inputs from OOD inputs. Benchmarks over many medical datasets report that no single static score works across semantic, covariate and far shifts [5, 9].

The softmax over C class prompts compresses a d-dimensional image embedding into C numbers, and a shifted image that the model classifies confidently is indistinguishable from an ID image in that view. Static scores read only this view, whether they take the maximum probability (MCM [19]), the maximum logit or the energy [6, 17], add patch-level matches (GL-MCM [21]), or average over augmented prompts (TAG [18]). Hierarchical prompts enrich the text side and improve detection on medical VLMs, at the price of expert-written prompt trees [9]. Few-shot methods learn prompts or alignments from labelled ID

images [14, 20, 27, 31]. Test-time methods adapt a VLM from the unlabelled stream itself, by learning textual OOD concepts (TTL [30]), negative semantics (TINS [29]) or prompts from a memory of confident samples (STAR [4]), and they update text embeddings or prompts. PROTON [2] is the closest work: it keeps the VLM frozen, maintains an online bank of class prototypes from confident predictions, and fuses the distance to the nearest prototype with MCM. Its evidence is a single ophthalmology benchmark, and whether the mechanism survives an independent reproduction is open.

Shifted inputs occupy regions of the embedding space that ID inputs do not, while their MCM scores overlap with ID scores, as Figure 1 shows for three modalities. The prototype bank therefore reads the right signal, but the confidence gate is the wrong filter. In our reproduction 97% of the ultra-widefield images pass the gate, 148 of 150 are assigned to the diabetic-retinopathy class, and that class bank ends with 63% shifted members, so the prototype moves towards the shift it should detect. A stricter gate does not help, because the shifted images are as confident as the ID images. PROTON+ replaces the filter with two label-free statistics of the bank itself. A mean-shift step over all bank members finds the dominant imaging-domain mode, and each class prototype is the mean of its members weighted by their proximity to that mode, so a confidently classified but shifted cluster receives almost no weight. A sample enters the bank only if its distance to the prototype of its predicted class is typical of the samples admitted before it. The two scores are fused as online z-scores, which removes the scale mismatch between a probability and a cosine distance.

We evaluate on three benchmarks with semantic, covariate and far-OOD splits: retinal photographs with `FLAIR`, chest radiographs with `UniMed-CLIP`, and histopathology patches with `QuiltNet`, against seven zero-shot scores and two detectors that use labelled ID features. PROTON+ improves over MCM on covariate and far shifts in every modality, and on semantic shift in radiology and pathology. On the ultra-widefield covariate shift, where PROTON loses 22.1 AUROC points against MCM, PROTON+ gains 16.4 points. The prototype distance alone, computed from the PROTON+ bank, reaches 90 to 99 AUROC on covariate and far shifts, and fusing it with a weak base score is what caps the fused detector. Both PROTON and PROTON+ assume that ID inputs form the majority of the stream, and their scores invert when shifted inputs make up half of it or more.

Our contributions are:

1. We reproduce PROTON from scratch on three modalities and release the library, manifests and cached features. The reported gains do not transfer: on the ultra-widefield covariate shift PROTON scores 22.1 AUROC points below MCM.

2. We trace the failure to bank contamination by confidently misclassified shifted inputs, and show that the variance-driven fusion weight stays between 0.42 and 0.65 on every stream, so it is a constant in practice.

3. We introduce PROTON+, which anchors prototypes to the dominant imaging-domain mode, gates bank admission by prototype consistency, and fuses z-scores, at 0.03 ms per sample and 0.4 MB of state.

4. We quantify the ceiling of the prototype signal with an oracle bank that never admits a shifted sample, and the sensitivity of every detector to the gate, the bandwidth and the prompt wording, which moves MCM by up to 20 AUROC points.

2 Related Work

Zero-shot OOD scores for vision-language models. MCM scores an image by the maximum softmax probability over the class prompts, scaled by the temperature of the contrastive model [19]. Maximum-logit and energy scores read the unnormalised similarities instead of the probabilities [6, 17]. GL-MCM

adds the best match between any image patch and any class prompt [21], and TAG averages the maximum probability over randomly shuffled prompt wordings [18]. All of these are stateless, so two images with the same probabilities receive the same score regardless of where they lie in the embedding space [28].

OOD detection for medical vision-language models. Ju et al. build a cross-modality benchmark with semantic, covariate and far shifts for `FLAIR`, `UniMed-CLIP` and `QuiltNet`, show that no static score wins on all shifts, and propose hierarchical prompts that describe anatomy, modality and lesion morphology [9]. OpenMIBOOD reaches the same conclusion over fourteen datasets [5]. Hierarchical vision-language learning and global-local alignment improve medical OOD detection with labelled ID images [14, 27]. Our work uses the ophthalmology benchmark of Ju et al. and builds the radiology and pathology benchmarks from public datasets with the same three shift types.

Test-time adaptation and memory banks. Prompt learning adapts a VLM from a few labelled images [31], and LoCoOp regularises the prompts with local background features for OOD detection [20]. Test-time methods drop the labels and learn from the unlabelled stream: TTL updates textual prompts with pseudo-labelled test samples and a bank of textual features [30], TINS learns negative text embeddings that are kept apart from ID prototypes [29], and STAR stores confident samples in a memory bank to build class prototypes for prompt adaptation [4]. These methods update text embeddings or prompts and target the accuracy of the classifier on the shifted inputs. PROTON keeps the model frozen and uses an online bank of image prototypes only to score inputs [2]. Our work keeps the bank and the fusion of PROTON and changes how prototypes are estimated and which samples may enter the bank.

Feature-bank detectors with labels. Nearest-neighbour and Mahalanobis distances to labelled ID features are strong detectors when such features are available [15, 24]. We report them as an upper bound that none of the zero-shot methods may use.

3 Method

PROTON+ scores every image of an unlabelled stream with a frozen VLM, an online bank of class prototypes, and a fusion of a softmax score with a prototype distance. We first state the setting and PROTON, then show why its bank gets contaminated, and then describe the three changes of PROTON+: domain-anchored prototypes, consistency admission and scale-free fusion.

3.1 Setting and PROTON

A frozen VLM maps an image x_t to an ℓ_2-normalised embedding $e_t \in \mathbb{R}^d$ and the C class names to text embeddings $T = [t_1, \dots, t_C]$, computed once. The zero-shot classifier is $p_t = \mathrm{softmax}(e_t^\top T/\tau)$ with the temperature τ learned by the VLM, and MCM scores $S_{\mathrm{MCM}}(x_t) = -\max_c p_{t,c}$, so that a higher score means more OOD. Images arrive one at a time, no labels and no training data are available, and the detector may use only the images it has already seen.

PROTON [2] keeps one first-in-first-out queue $\mathcal{D}_c$ of at most M embeddings per class. After an image is scored, it enters the queue of its predicted class $\hat{c} = \arg\max_c p_{t,c}$ if $\max_c p_{t,c} \geq \gamma$, and the oldest member leaves when the queue is full. The prototype of class c is the normalised mean of its queue,

$$\mu_c = \bar{e}_c / \|\bar{e}_c\|_2, \qquad \bar{e}_c = \frac{1}{|\mathcal{D}_c|} \sum_{e \in \mathcal{D}_c} e, \tag{1}$$

and the prototype distance is one minus the cosine similarity to the nearest prototype,

$$S_{\mathrm{proto}}(x_t) = 1 - \max_c e_t^\top \mu_c. \tag{2}$$

Once every queue holds at least $K_{\min}$ members, the score is the convex combination

$$S_t = \alpha_t \, S_{\mathrm{MCM}}(x_t) + (1 - \alpha_t) \, S_{\mathrm{proto}}(x_t), \qquad \alpha_t = \alpha_{\max} - \sigma\big(100 \, (\sigma_t^2 - \sigma_0^2)\big) \, (\alpha_{\max} - \alpha_{\min}), \quad (3)$$

where σ_t^2 is the running variance of S_{MCM} over the stream, maintained with Welford's algorithm [26], σ_0^2 is a reference variance, $\sigma(\cdot)$ is the logistic function, and α_t moves from $\alpha_{\max}$ towards $\alpha_{\min}$ when the variance of S_{MCM} rises above the reference. Before calibration the score is S_{MCM} alone.

3.2 Why the bank gets contaminated

The gate $\max_c p_{t,c} \geq \gamma$ is meant to keep OOD embeddings out of the bank. It can only exclude inputs with a low maximum probability, which are exactly the inputs that MCM already detects. A covariate-shifted input is one that the model classifies confidently for the wrong reason, so it passes the gate by definition and lands in the bank of the class it was assigned to. When such inputs are a sizeable share of the stream, the prototype of that class becomes the mean of two clusters, the ID cluster and the shifted cluster, and lies between them. Every member of either cluster is then at a similar distance from the prototype, and Equation (2) no longer separates them. The fusion in Equation (3) adds a second problem, because S_{MCM} lies in $[-1, -1/C]$ while S_{proto} lies in $[0, 2]$, so the weight α_t does not control the relative influence of the two signals.

3.3 PROTON+

Domain-anchored prototypes. All ID classes of a benchmark share one imaging domain, so the union of all queues contains one dominant mode, and a shifted cluster inside a single queue is a minority of the union as long as ID inputs form the majority of the stream. A mean-shift step [1] over the union finds that mode. Starting from the normalised mean μ of all members $\mathcal{D} = \bigcup_c \mathcal{D}_c$, we iterate

$$\mu \leftarrow \frac{\sum_{e \in \mathcal{D}} w(e, \mu) \, e}{\big\| \sum_{e \in \mathcal{D}} w(e, \mu) \, e \big\|_2}, \qquad w(e, \mu) = \exp\big((e^\top \mu - 1)/h\big), \quad (4)$$

three times, where h is a bandwidth on the cosine scale and w down-weights members that are far from the current mode. The result μ_{dom} is the domain anchor. Each class prototype is then the domain-weighted mean of its own queue,

$$\mu_c = \frac{\sum_{e \in \mathcal{D}_c} w(e, \mu_{\mathrm{dom}}) \, e}{\big\| \sum_{e \in \mathcal{D}_c} w(e, \mu_{\mathrm{dom}}) \, e \big\|_2}, \quad (5)$$

so a member whose cosine similarity to the anchor is lower by h than that of a typical ID member contributes e^{-1} as much, and a member in a shifted cluster at cosine distance several times h contributes almost nothing. The prototype distance is still Equation (2).

Consistency admission. After calibration a sample enters the queue of its predicted class $\hat{c}$ only if its confidence passes the gate and its distance $d_t = 1 - e_t^\top \mu_{\hat{c}}$ is typical of the samples admitted before it,

$$d_t \leq m_t + \kappa \, s_t, \quad (6)$$

where m_t and s_t are the running mean and standard deviation of the distances of admitted samples, also maintained with Welford's algorithm, and κ sets the tolerance. The rule costs one dot product and three scalar updates, and it rejects the members that Equation (5) would have down-weighted, so the queue fills with ID samples and the FIFO eviction removes early contamination over time.

Scale-free fusion. We keep Equation (3) with the same weight α_t but replace each score by its online z-score, $(S_{\mathrm{MCM}} - \bar{m}_t)/\hat{s}_t$ and $(S_{\mathrm{proto}} - \bar{p}_t)/\hat{q}_t$, where the means and standard deviations are those of the scores seen so far in the stream. Before calibration the z-scored MCM is emitted, so the score keeps one

scale across the calibration boundary. A variant takes the maximum of the two z-scores instead of their weighted sum, which flags an input when either signal is anomalous and has no fusion weight.

Cost. Per sample PROTON+ computes C dot products for the prototype distance, one for the admission rule, and after each admission a mean-shift step over at most CM members, which is three passes over a $CM \times d$ matrix. The state is the bank of CM embeddings and a handful of scalars, 0.4 MB for $C = 2$, $M = 100$ and $d = 512$. No gradient, augmentation or text re-encoding is involved, and any score from Section 2 can replace MCM as the base signal.

4 Experiments

Benchmarks. We use three benchmarks, each with two ID classes and three OOD splits, and we follow the construction of Ju et al. where it exists [9]. *Ophthalmology* uses FIVES [8] with normal and diabetic retinopathy as ID classes (150 images each), glaucoma and age-related macular degeneration from the same dataset as the semantic shift (150 each), ultra-widefield fundus photographs from DeepDRiD [16] as the covariate shift (150), and natural images from ImageNet [3] as the far shift (150). *Radiology* uses the RSNA pneumonia challenge [22] with normal and lung-opacity radiographs as ID classes (300 each), NIH ChestX-ray14 radiographs whose findings lie outside the ID classes as the semantic shift (300), paediatric radiographs [11] of both classes as the covariate shift (150), and ImageNet as the far shift (150). *Pathology* uses PatchCamelyon [25] with benign and malignant lymph-node patches as ID classes (300 each), five colorectal tissue types from NCT-CRC-HE [10] with no counterpart in the ID classes as the semantic shift (300), patches of both classes from the held-out hospital of Camelyon17-WILDS [13] as the covariate shift (150), and ImageNet as the far shift (150). Every image list is sampled once with a fixed seed and released with the code. A disjoint labelled ID split of the same size as the ID test set serves only the two detectors that need labelled features.

Models and prompts. The frozen VLMs are `FLAIR` (ResNet-50 and Bio_ClinicalBERT, $d = 512$) [23], `UniMed-CLIP` (ViT-B/16 and BiomedBERT) [12] and `QuiltNet` (ViT-B/32) [7], with the temperature each model learned. Prompts follow the public benchmark code: `FLAIR` uses its built-in caption "A fundus photograph of {class}" and `UniMed-CLIP` uses "x-ray of {class}". No benchmark code exists for PatchCamelyon, so for `QuiltNet` we chose "normal" and "cancer" among twelve wordings by zero-shot accuracy on the disjoint labelled split, which never touches the evaluated images. `FLAIR` averages text-encoder states over all tokens of a padded batch, so the class prompts are encoded as one batch, exactly as its forward pass does. The zero-shot accuracies on the ID test images are 84.0%, 69.2% and 72.7%.

Streams and protocol. A stream contains all ID test images and all images of one OOD split in a random order, so the ID fraction is 0.5 for the semantic split and 0.67 for the other two. Every online detector is run over five random orders, and all numbers for online detectors are the mean $\pm$ standard deviation over these five orders. Static scores do not depend on the order. We report AUROC, the area under the precision-recall curve with OOD as the positive class (AUPR), and the false-positive rate on ID images at 95% OOD recall (FPR@95), all in percent. PROTON runs with the values of its paper, $\gamma = 0.7$, $M = 100$, $K_{\min} = 5$, $\alpha_{\min} = 0.3$, $\alpha_{\max} = 0.7$ and $\sigma_0^2 = 0.02$. PROTON+ adds $h = 0.1$, $\kappa = 2$ and three mean-shift iterations, set before any experiment from the typical within-class cosine spread and the two-sigma rule, and never tuned. Features are extracted once per benchmark on one GPU in under a minute, and every detector then runs on a CPU.

Baselines. The zero-shot baselines are Max-Logits, Energy, MCM, GL-MCM, TAG with ten shuffled prompt wordings, and the hierarchical prompts of Ju et al. at one and five descriptions per class. Entropy equals MCM for two classes and is omitted. kNN and Mahalanobis distances to the labelled ID split are reported as privileged upper bounds. The following subsections ask whether PROTON reproduces, why it fails, what PROTON+ adds, which of its parts matter, how sensitive it is, and how it behaves when shifted

Table 1: **PROTON does not reproduce on the ophthalmology benchmark, PROTON+ recovers covariate and far shifts.** Zero-shot OOD detection with `FLAIR`. AUROC and AUPR higher is better, FPR@95 lower is better. Online detectors report mean ± SD over five stream orders. † hierarchical prompts of Ju et al. Best per column in bold, second best underlined.

	Semantic			Covariate			Far-OOD		
Method	AUC↑	APR↑	FPR↓	AUC↑	APR↑	FPR↓	AUC↑	APR↑	FPR↓
Max-Logits	54.0	54.1	<u>79.3</u>	27.9	23.3	88.0	83.9	76.4	63.0
Energy	51.8	51.5	79.7	27.2	23.2	88.0	83.9	<u>76.5</u>	62.7
GL-MCM	**75.8**	**75.6**	86.0	68.9	45.3	69.3	78.5	62.4	75.0
MCM	<u>75.8</u>	<u>75.6</u>	86.0	68.9	45.2	69.7	78.5	62.4	75.0
TAG-MCM	68.4	70.5	88.7	30.1	23.8	93.7	72.2	47.2	60.3
MCM (L=1)	66.5	65.0	84.7	<u>84.2</u>	**66.3**	<u>47.3</u>	51.8	38.4	98.3
MCM (L=5)	58.3	56.5	87.0	66.5	42.5	65.7	54.4	33.5	74.3
PROTON [2]	68.0	72.5	92.7	46.7	29.4	83.6	<u>86.8</u>	76.2	<u>55.0</u>
PROTON+ (ours)	67.3	69.0	**75.8**	**85.3**	<u>63.9</u>	**31.5**	**93.4**	**81.8**	**15.4**
Δ PROTON+ vs. MCM	▼8.4	▼6.5	▲10.2	▲16.4	▲18.6	▲38.1	▲14.9	▲19.4	▲59.6

inputs dominate the stream.

4.1 Does PROTON reproduce?

Table 1 lists every zero-shot score on the ophthalmology benchmark, with the AUROC, AUPR and FPR@95 of each shift, and Table 2 gives the three modalities. A row above the MCM row on AUROC beats the static baseline the online detectors build on. **PROTON improves on far shifts in every modality and on the covariate shift in radiology and pathology, but it loses** 22.1 **AUROC points to MCM on the ultra-widefield covariate shift and** 7.7 **points on the semantic shift of the same benchmark.** The running variance of MCM stays below 0.03 on every stream, so with $\sigma_0^2 = 0.02$ the fusion weight ranges only from 0.42 to 0.65 and the adaptive fusion behaves as a constant. The confidence gate costs rather than helps in radiology and pathology, where the bank without a gate scores about 5 points higher on covariate shift, as Table 3 shows for ophthalmology and the appendix for the other two modalities. The covariate gains in radiology (+14.6) and pathology (+7.6) are far below the gains the original paper reports, and the order of the stream changes the AUROC by 1 to 4 points.

4.2 Why does it fail? Bank contamination

Figure 1 plots, per modality, the overlap of MCM scores between ID and OOD, the MCM confidence against the prototype distance, and a t-SNE of the embeddings. Covariate and far samples form clusters away from the ID classes in every modality, while their MCM scores overlap with ID scores. The gate admits what it should exclude. On the ultra-widefield stream 97% of the shifted images pass $\gamma = 0.7$ with a mean confidence of 0.955 against 0.970 for ID images, 148 of 150 are assigned to the diabetic-retinopathy class, and that queue ends with 63% shifted members. On the paediatric radiograph stream the normal queue ends with 59% shifted members, and on the pathology streams the shifted share of a queue reaches 54%. **With an oracle gate that never admits a shifted sample, PROTON separates the ultra-widefield images from**

Table 2: **PROTON+ improves over MCM on covariate and far shifts in every modality.** AUROC, AUPR and FPR@95 per shift for the three benchmarks, mean ± SD over five stream orders for the online detectors.

		Semantic			Covariate			Far-OOD		
Domain (VLM)	Method	AUC	APR	FPR	AUC	APR	FPR	AUC	APR	FPR
Ophthalmology (FLAIR)	MCM	75.8	75.6	86.0	68.9	45.2	69.7	78.5	62.4	75.0
	PROTON	68.0 ± 0.9	72.5 ± 0.7	92.7 ± 2.3	46.7 ± 1.8	29.4 ± 0.8	83.6 ± 6.1	86.8 ± 3.6	76.2 ± 2.6	55.0 ± 26.0
	PROTON+	67.3 ± 0.7	69.0 ± 0.3	75.8 ± 1.7	85.3 ± 7.8	63.9 ± 6.8	31.5 ± 26.5	93.4 ± 0.5	81.8 ± 1.2	15.4 ± 1.8
Radiology (UniMed-CLIP)	MCM	56.8	37.1	87.3	46.6	18.1	87.3	27.6	14.0	99.7
	PROTON	57.1 ± 1.2	39.6 ± 0.9	94.4 ± 1.1	61.2 ± 1.7	31.0 ± 1.4	67.6 ± 3.7	93.9 ± 2.3	92.5 ± 2.1	33.5 ± 33.5
	PROTON+	65.2 ± 0.5	45.3 ± 1.1	85.8 ± 2.7	93.9 ± 0.6	82.1 ± 3.4	26.7 ± 5.4	91.5 ± 1.4	74.6 ± 4.4	35.7 ± 31.5
Pathology (QuiltNet)	MCM	48.8	33.1	97.2	48.0	19.4	92.0	45.1	18.7	95.0
	PROTON	52.9 ± 0.8	36.5 ± 0.8	94.0 ± 1.3	55.6 ± 1.4	28.7 ± 1.4	90.3 ± 3.6	58.2 ± 2.9	33.1 ± 3.6	89.2 ± 5.3
	PROTON+	72.5 ± 2.0	57.8 ± 2.9	73.8 ± 2.7	70.1 ± 1.6	44.1 ± 1.7	76.5 ± 1.9	81.1 ± 1.1	61.9 ± 1.4	64.3 ± 4.5

ID at 93.1 AUROC instead of 46.7, so the signal is there and the filter is the fault.

4.3 What does PROTON+ add?

Table 2 compares MCM, PROTON and PROTON+ on the three modalities. **PROTON+ beats MCM on the covariate and far shifts of every modality, by 16.4 to 47.3 AUROC points on covariate shift and by 14.9 to 63.9 points on far shift, and it beats PROTON on seven of the nine shift-modality pairs, trailing by 0.7 points on the ophthalmology semantic shift and by 2.4 points on the radiology far shift.** On semantic shift it gains 8.4 points in radiology and 23.7 points in pathology, and loses 8.5 points in ophthalmology, where the glaucoma and macular-degeneration images lie inside the ID imaging domain and MCM already separates them at 75.8. FPR@95 on the ultra-widefield shift drops from 69.7 for MCM and 83.6 for PROTON to 31.5, and on the paediatric shift from 87.3 and 67.6 to 26.7. The standard deviation over stream orders is below 2 points everywhere except the ultra-widefield shift (7.8), where the first 14 samples decide how much of the shifted cluster is admitted before the anchor forms.

4.4 Which parts matter?

Table 3 adds the three changes one at a time on the ophthalmology benchmark, and Table 4 lists the prototype score on its own and the oracle bank for every modality. Z-scoring alone lifts covariate shift by 4 to 17 points, and far shift by 2 and 11 points in ophthalmology and pathology at a cost of 3 points in radiology, because the raw MCM score spans half a unit while the raw prototype distance spans two. Consistency admission alone cannot repair a bank whose seed is already mixed, and on the ultra-widefield shift it lowers the AUROC to 40.1. **Domain-anchored prototypes carry the covariate gain: they raise the ultra-widefield shift from 40.1 to 85.3 and the paediatric shift from 78.4 to 93.9.** The prototype distance computed from the PROTON+ bank is the strongest single signal on covariate and far shifts, with 90.4, 98.8 and 90.1 AUROC on the three covariate shifts and 96 to 98 on the far shifts, and it beats the fused score in pathology on every shift. Fusion with MCM pays only where MCM carries information the embedding lacks, which is the ophthalmology semantic shift (57.0 alone against 67.3 fused). The oracle bank bounds what any admission rule can achieve: PROTON+ with an oracle gate reaches 73.6, 90.4 and 93.5 in ophthalmology, so the remaining gap of PROTON+ to its oracle is 6.3 points on semantic and 5.1 points on covariate shift. The maximum of the two z-scores replaces the fusion weight and gains 5 to 14 points in pathology, where MCM is near chance, and changes the other six pairs by -6 to $+1$ points.

Table 3: **Domain anchoring carries the covariate gain.** Ablation on the ophthalmology benchmark (AUROC). The top block adds the components of PROTON as in its paper, the bottom block adds the changes of PROTON+ one at a time.

Variant	Sem	Cov	Far
MCM baseline	75.8	68.9	78.5
+ Proto. (α=.5)	65.5	46.6	86.9
+ Adapt. fusion	65.3	46.9	85.3
+ Conf. gate (Full)	68.0	46.7	86.8
PROTON + z-scored fusion	69.4	50.7	89.0
+ consistency admission	70.6	40.1	92.6
+ domain-anchored prototypes (PROTON+)	67.3	85.3	93.4

Table 4: **The prototype signal, not the fusion, is the detector.** AUROC of the prototype distance alone from the PROTON+ bank, and of PROTON+ with an oracle gate that never admits a shifted sample.

Modality	Variant	Sem	Cov	Far
Ophth.	PROTON+	67.3	85.3	93.4
	S_{proto} only (PROTON+ bank)	57.0	90.4	97.9
	PROTON+, oracle bank	73.6	90.4	93.5
Radio.	PROTON+	65.2	93.9	91.5
	S_{proto} only (PROTON+ bank)	63.6	98.8	96.2
	PROTON+, oracle bank	68.1	92.4	91.3
Patho.	PROTON+	72.5	70.1	81.1
	S_{proto} only (PROTON+ bank)	91.0	90.1	98.2
	PROTON+, oracle bank	76.2	72.0	79.6

Table 5: **PROTON is stable in M but its covariate AUROC rises with a stricter gate, because fewer shifted images are admitted.** Ultra-widefield covariate shift, PROTON with the defaults of its paper otherwise.

γ	AUC	APR	F95	M	AUC	APR	F95
0.5	46.9	29.4	82.5	10	45.3	29.3	92.0
0.6	46.7	29.4	83.8	25	44.6	28.6	89.9
0.7	46.7	29.4	83.6	50	45.4	28.9	87.2
0.8	47.8	30.1	81.3	100	46.7	29.4	83.6
0.9	49.4	30.7	79.9	200	47.9	29.9	79.5

4.5 How sensitive is it?

Table 5 sweeps the gate γ and the bank size M for PROTON, and the appendix sweeps γ, κ, h and the prompt wording for PROTON+. **PROTON+ is flat in κ (85.0 to 87.7 on the ultra-widefield shift for $\kappa \in [1, 4]$) and in $\gamma \in [0.5, 0.7]$ in every modality.** A stricter gate starves the bank where the ID images are less confident: with mean ID confidences of 0.82 in radiology and 0.67 in pathology, $\gamma = 0.8$ costs 5 to 13 points in pathology, and $\gamma = 0.9$ drops the pathology shifts to chance and the radiology far shift by 20 points. A bandwidth of $h \leq 0.1$ works in every modality, $h = 0.2$ loses 13 points on the ultra-widefield shift and 12 points on the pathology semantic shift, and $h = 0.5$ no longer separates the ultra-widefield cluster from the ID mode. Prompt wording moves the static baseline more than any detector choice: a trailing period in the `FLAIR` prompt moves the MCM covariate AUROC from 68.9 to 84.2 and the far AUROC from 78.5 to 51.8, while PROTON+ stays between 85.3 and 94.0 on covariate and between 88.1 and 97.0 on far shift across the three single-prompt wordings. Any base score benefits from the bank: with Max-Logits or Energy, which score 27 to 28 on the ultra-widefield shift, PROTON+ reaches 87 on covariate and 99 on far shift.

4.6 When shifted inputs dominate the stream

Table 6 varies the ID fraction ρ of the stream by sub-sampling the smaller pool. **Both detectors assume an ID majority: at $\rho = 0.9$ and 0.7 PROTON+ beats MCM on covariate and far shifts in every modality, and once the shifted cluster dominates the bank the prototype score inverts.** The inversion starts at $\rho = 0.5$ in ophthalmology and pathology and at $\rho = 0.3$ in radiology, where the ultra-widefield AUROC of

Table 6: **Online prototypes need an ID majority.** AUROC against the ID fraction ρ of the ophthalmology stream (Sem / Cov / Far).

	MCM			PROTON			PROTON+		
ρ	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
90%	76.2	67.7	75.5	73.1	85.5	90.4	70.6	88.5	90.9
70%	74.8	69.0	79.0	71.2	47.7	87.4	68.7	74.7	92.9
50%	75.8	69.1	78.4	68.2	34.0	55.3	67.4	22.3	36.7
30%	76.2	68.3	78.0	60.3	28.0	26.4	64.5	28.8	27.8
10%	75.9	69.2	78.1	50.0	41.3	16.7	56.8	41.4	20.5

PROTON+ is 22.3 at $\rho = 0.5$ and the far-OOD AUROC is 20.5 at $\rho = 0.1$, against 69 and 78 for MCM, which is order-free. PROTON degrades in the same way, with 34.0 and 16.7 at the same points, so the claim of its paper that it never falls below MCM does not hold for streams with an ID minority.

4.7 Cost

One PROTON+ step takes 0.02 to 0.03 ms on a single CPU core in NumPy, including the mean-shift step after an admission, and the state is 0.41 MB for $C = 2$, $M = 100$ and $d = 512$. The detector is calibrated after 14, 30 and 36 samples on the three benchmarks, which is when every queue holds $K_{\min} = 5$ members.

5 Discussion

The prototype signal is the detector. On every covariate and far shift the prototype distance from the PROTON+ bank beats the fused score, and on seven of nine pairs it beats every static baseline. The fusion of the original design protects the semantic shifts of a modality whose embedding keeps the shifted classes inside the ID domain, as in ophthalmology, and costs up to 20 points elsewhere. Choosing the weight from the stream without labels is the open question: the variance of MCM does not indicate a shift, and we found no single unsupervised statistic of the two score distributions that picks the better signal on all nine streams.

Towards streams with an ID minority. Both detectors invert once the shifted cluster dominates the bank, because the dominant mode is then the shift. The text prompts are the only label-free anchor of the ID domain, and in our data the shifted images are as close to the prompts as the ID images, so a text-anchored mode would not help. A deployment that cannot guarantee an ID majority needs a small labelled seed set, which turns the bank into the nearest-neighbour detector of the appendix.

Towards shift-aware benchmarks. Every OOD split in the three benchmarks, and in the benchmark of Ju et al., comes from a different dataset than the ID split, and the privileged detectors separate them at 99 to 100 AUROC. The benchmarks therefore measure how well a detector recovers dataset identity from the embedding without labels, which is the quantity test-time prototypes target, and they do not isolate a shift type from an acquisition difference. Splits drawn from the same acquisition pipeline, as our radiology semantic split from NIH ChestX-ray14, are the step towards that.

Towards robust prompts. A trailing period moves the MCM covariate AUROC of `FLAIR` by 15 points, and the text embedding of a prompt depends on the other prompts in its batch. Any comparison of zero-shot detectors on medical VLMs should report the prompt wording and the encoding protocol, and PROTON+ reduces but does not remove this dependence.

6 Conclusion

We reproduce PROTON, an online prototype-bank detector for medical vision-language models, on three modalities and find that its confidence gate admits the confidently misclassified inputs it is meant to exclude, so the prototypes absorb the shift. PROTON+ anchors the prototypes to the dominant imaging-domain mode of the bank, admits only samples consistent with their class prototype, and fuses z-scores. With the same hyper-parameters it improves over MCM on covariate and far shifts in every modality, by up to 47.3 and 63.9 AUROC points, and closes most of the gap to an oracle bank. The library, the image manifests, the cached features and a browser demo with the real `FLAIR` weights are released so that every number in this paper can be recomputed.

AI assistance. An AI coding assistant (Claude) implemented the library, ran the experiments and drafted the text under the authors' direction, and the authors reviewed the code, the protocol and every number.

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A Benchmark construction

Ophthalmology. FIVES holds 800 fundus photographs of 2048 by 2048 pixels, 200 per class, split by the dataset into training and test folders that we merge. Per ID class we draw 150 images for the test stream and keep the remaining 50 as the labelled split, and per semantic class we draw 150. The 150 ultra-widefield images are drawn from the 256 photographs of the three DeepDRiD ultra-widefield folders. ImageNet images are 150 validation images of the ten-class Imagenette subset at full resolution.

Radiology. The RSNA images are the DICOM radiographs of the challenge training set, converted to 8-bit PNG with per-image min-max scaling. We draw 600 per class and use the first 300 as the test stream and the rest as the labelled split. The semantic split is drawn from the public 5,606-image sample of NIH ChestX-ray14 after removing every image whose findings include No Finding, Pneumonia, Infiltration, Consolidation, Effusion or Edema, so that cardiomegaly, pneumothorax, masses, nodules, atelectasis and fibrosis remain. The paediatric split draws 75 normal and 75 pneumonia images from the test folder of Kermany et al. A second semantic split of 300 COVID-19 radiographs from the COVID-19 Radiography Database is released with the manifests but not used in the tables, because COVID-19 pneumonia overlaps the pneumonia class.

Pathology. PatchCamelyon patches are 96 by 96 pixels. We draw 600 per label from the test split, 300 for the stream and 300 for the labelled split. The semantic split draws 60 patches from each of adipose, debris, mucus, smooth muscle and cancer-associated stroma in NCT-CRC-HE-7K. Normal mucosa, lymphocytes, tumour epithelium and background are excluded because they resemble the ID classes or carry no tissue. The covariate split draws 75 patches per label from the out-of-distribution test hospital of Camelyon17-WILDS.

B Prompt sets

Tables 7 to 9 report every prompt set with its zero-shot ID accuracy and the AUROC of MCM, PROTON and PROTON+. “vanilla” is the bare class name, “l1” the one-description template of Ju et al. (for `FLAIR` the built-in caption with a trailing period, for `UniMed-CLIP` the benchmark prompt, for `QuiltNet` a

histopathology template), and "l5" the five-description hierarchy of Ju et al., which we wrote for radiology and pathology in the same style.

Table 7: Prompt sensitivity, ophthalmology (AUROC Sem / Cov / Far).

Prompt set	ID acc.	MCM (Sem/Cov/Far)	PROTON	PROTON+
flair_default	84.0	75.8 / 68.9 / 78.5	68.0 / 46.7 / 86.8	67.3 / 85.3 / 93.4
vanilla	86.0	65.6 / 64.9 / 83.5	58.5 / 46.7 / 94.4	58.2 / 87.9 / 97.0
l1	83.7	66.5 / 84.2 / 51.8	65.0 / 66.6 / 78.2	64.6 / 94.0 / 88.1
l5	57.3	58.3 / 66.5 / 54.4	41.2 / 24.0 / 25.3	56.2 / 50.8 / 58.8

Table 8: Prompt sensitivity, radiology.

Prompt set	ID acc.	MCM (Sem/Cov/Far)	PROTON	PROTON+
vanilla	73.2	38.6 / 57.5 / 55.5	42.5 / 78.9 / 94.7	51.8 / 95.5 / 96.2
l1	69.2	56.8 / 46.6 / 27.6	57.1 / 61.2 / 93.9	65.2 / 93.9 / 91.5
l5	55.3	56.1 / 78.4 / 84.6	52.1 / 79.1 / 88.3	52.7 / 88.5 / 96.8

Table 9: Prompt sensitivity, pathology.

Prompt set	ID acc.	MCM (Sem/Cov/Far)	PROTON	PROTON+
id_acc	72.7	48.8 / 48.0 / 45.1	52.9 / 55.6 / 58.2	72.5 / 70.1 / 81.1
vanilla	64.5	64.3 / 53.4 / 72.5	66.7 / 58.9 / 82.2	75.8 / 72.9 / 92.5
l1	59.3	59.8 / 66.1 / 67.1	64.9 / 73.0 / 75.2	82.4 / 85.6 / 92.1
l5	65.5	15.3 / 64.6 / 61.1	32.6 / 61.5 / 65.7	15.4 / 73.4 / 81.5

C Full tables for radiology and pathology

D Broader baselines and base scores

Table 15 lists TAG, the five-description hierarchy, the two privileged detectors and PROTON+, and Table 16 instantiates PROTON and PROTON+ on six base scores. The privileged detectors use the labelled split and separate every OOD split at 99 to 100 AUROC on covariate and far shifts. The hierarchy of Ju et al. with PROTON on top inverts in ophthalmology, because the five-description prompts classify the ID images at 57% accuracy and the bank fills with misrouted samples.

E Sensitivity of PROTON+

Table 17 sweeps the mean-shift bandwidth h, and Figure 2 the gate and bank size of PROTON on all three modalities. The reference variance σ_0^2 of the fusion weight is not inert: sweeping it from 0.005 to 0.08

Table 10: Zero-shot OOD detection with `UniMed-CLIP` on the radiology benchmark.

	Semantic			Covariate			Far-OOD		
Method	AUC↑	APR↑	FPR↓	AUC↑	APR↑	FPR↓	AUC↑	APR↑	FPR↓
Max-Logits	72.7	60.8	84.7	0.4	10.8	100.0	75.7	64.6	90.2
Energy	**73.1**	**61.9**	85.0	0.4	10.8	100.0	77.0	67.4	90.5
GL-MCM	56.8	37.6	87.3	46.7	18.1	87.3	29.2	14.2	98.0
MCM	56.8	37.1	87.3	46.6	18.1	87.3	27.6	14.0	99.7
TAG-MCM	60.1	39.3	83.7	48.1	19.3	85.5	41.3	16.0	87.5
MCM (L=1)	56.8	37.1	87.3	46.6	18.1	87.3	27.6	14.0	99.7
MCM (L=5)	56.1	33.1	**76.2**	78.4	38.2	46.3	84.6	52.3	39.5
PROTON [2]	57.1	39.6	94.4	61.2	31.0	67.6	**93.9**	**92.5**	**33.5**
PROTON+ (ours)	65.2	45.3	85.8	**93.9**	**82.1**	**26.7**	91.5	74.6	35.7
Δ PROTON+ vs. MCM	▲8.3	▲8.1	▲1.6	▲47.3	▲64.1	▲60.7	▲63.9	▲60.6	▲64.0

Table 11: Zero-shot OOD detection with `QuiltNet` on the pathology benchmark.

	Semantic			Covariate			Far-OOD		
Method	AUC↑	APR↑	FPR↓	AUC↑	APR↑	FPR↓	AUC↑	APR↑	FPR↓
Max-Logits	7.1	19.6	99.2	24.5	12.8	98.2	17.7	11.9	99.0
Energy	6.7	19.5	99.2	23.9	12.7	97.3	17.5	11.9	98.8
GL-MCM	46.5	31.2	97.7	46.0	18.1	96.8	44.2	18.4	96.8
MCM	48.8	33.1	97.2	48.0	19.4	92.0	45.1	18.7	95.0
TAG-MCM	48.7	33.1	97.2	48.0	19.4	92.0	45.1	18.7	95.0
MCM (L=1)	59.8	42.4	90.3	66.1	29.8	85.2	67.1	31.5	77.2
MCM (L=5)	15.3	20.4	99.8	64.6	30.7	77.3	61.1	26.3	87.3
PROTON [2]	52.9	36.5	94.0	55.6	28.7	90.3	58.2	33.1	89.2
PROTON+ (ours)	**72.5**	**57.8**	**73.8**	**70.1**	**44.1**	**76.5**	**81.1**	**61.9**	**64.3**
Δ PROTON+ vs. MCM	▲23.7	▲24.7	▲23.4	▲22.2	▲24.7	▲15.5	▲36.0	▲43.2	▲30.7

moves the mean AUROC of PROTON over the three shifts by 2 points in ophthalmology and by 14 points in radiology, in the direction of a smaller weight on MCM. A three-way triage of the shift type from the pair of scores reaches a macro-F1 of 0.60 in ophthalmology and 0.46 in pathology with a logistic regression under five-fold cross-validation.

F Text encoding in `FLAIR`

The text tower of `FLAIR` averages the hidden states of three BERT layers over all token positions of the padded batch, padding included. The embedding of "A fundus photograph of normal" therefore changes with the length of the longest prompt in its batch. The forward pass of the released model encodes all class prompts in one batch, and the benchmark notebooks of Ju et al. call that forward pass, so our main protocol encodes the class prompts jointly. Encoding each prompt alone, as the model's own helper for text embeddings does, moves the MCM covariate AUROC by up to 17 points.

Table 12: Ablation, radiology (AUROC).

Variant	Sem	Cov	Far
MCM baseline	56.8	46.6	27.6
+ Proto. (α=.5)	58.1	66.0	96.0
+ Adapt. fusion	58.1	67.7	97.2
+ Conf. gate (Full)	57.1	61.2	93.9
PROTON + z-scored fusion	65.1	78.1	90.5
+ consistency admission	65.5	78.4	92.8
+ domain-anchored prototypes (PROTON+)	65.2	93.9	91.5

Table 13: Ablation, pathology (AUROC).

Variant	Sem	Cov	Far
MCM baseline	48.8	48.0	45.1
+ Proto. (α=.5)	56.7	61.1	68.5
+ Adapt. fusion	54.4	57.8	62.5
+ Conf. gate (Full)	52.9	55.6	58.2
PROTON + z-scored fusion	59.6	64.5	69.2
+ consistency admission	53.8	65.9	70.9
+ domain-anchored prototypes (PROTON+)	72.5	70.1	81.1

Table 14: ID fraction sweep, radiology (top) and pathology (bottom), AUROC Sem / Cov / Far.

	MCM			PROTON			PROTON+		
ρ	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
90%	56.9	46.8	27.5	60.7	71.0	94.1	67.4	96.4	94.1
70%	57.4	47.2	27.9	58.6	54.3	81.8	65.6	89.5	81.0
50%	57.1	47.9	28.5	57.2	49.9	58.8	62.4	69.0	73.0
30%	57.2	46.7	27.7	56.0	38.8	37.1	58.8	12.2	14.2
10%	56.2	47.5	27.5	55.7	23.7	21.4	55.9	12.7	13.1

	MCM			PROTON			PROTON+		
ρ	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
90%	49.0	45.9	46.0	65.9	56.9	67.0	85.3	74.5	87.8
70%	48.9	48.5	45.6	54.7	50.7	52.7	72.5	62.5	73.8
50%	48.5	48.1	45.4	47.9	46.6	49.7	50.1	42.0	55.2
30%	49.4	48.7	45.7	45.2	40.3	43.8	32.1	27.9	39.1
10%	45.9	44.7	41.9	33.5	31.6	39.7	15.9	24.7	28.3

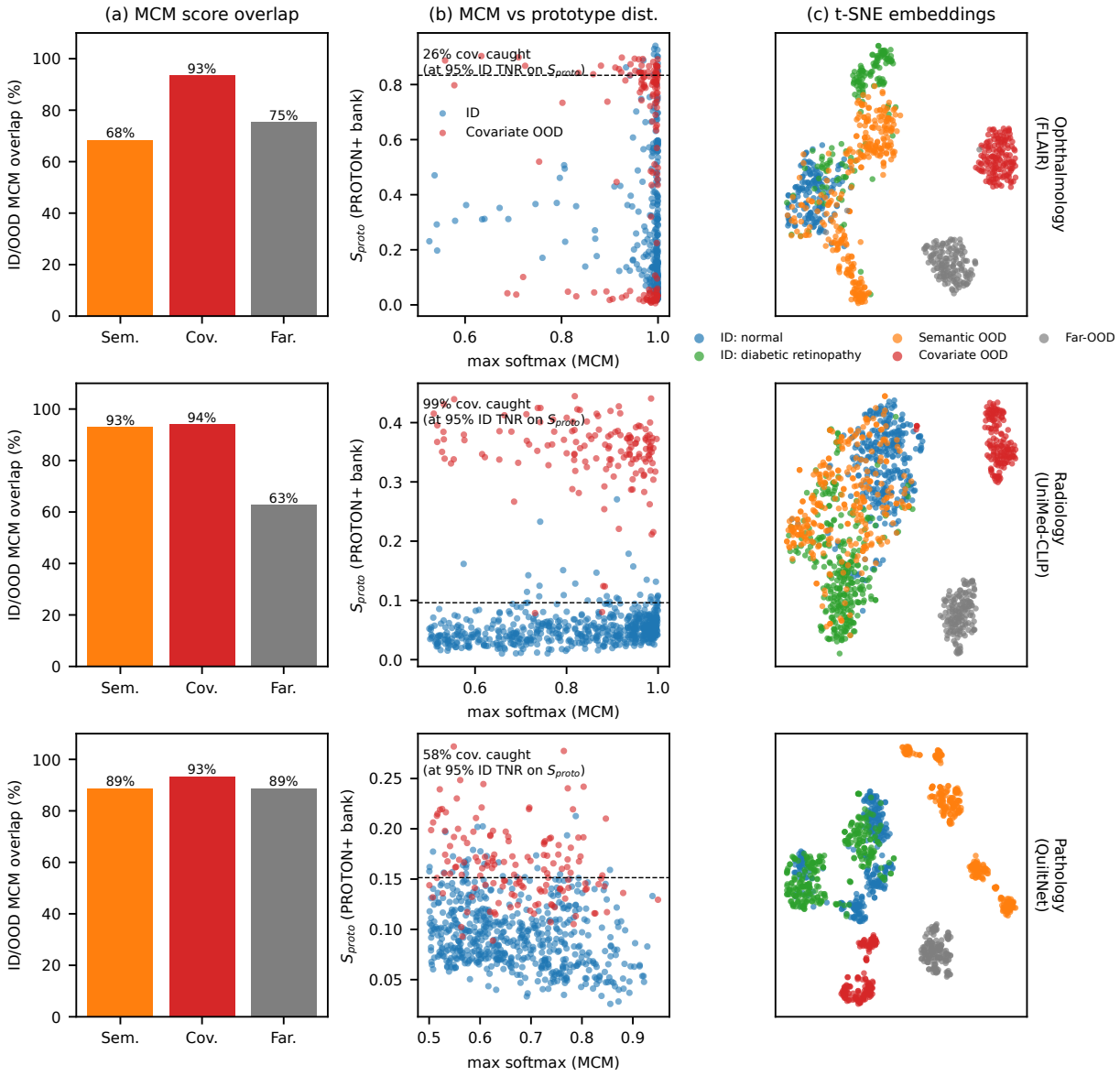


Figure 1: **Shifted inputs are separable in embedding space but not in softmax space.** (a) share of OOD MCM scores inside the 5th to 95th percentile range of ID scores, (b) MCM confidence against the online prototype distance on the covariate stream with the fraction of covariate samples above the 95th ID percentile, (c) t-SNE of the embeddings. Rows: `FLAIR`, `UniMed-CLIP`, `QuiltNet`.

Table 15: Broader baselines (AUROC Sem / Cov / Far). ‡ uses labelled ID features.

Ophthalmology

Method	Sem	Cov	Far
TAG-MCM	68.4	30.1	72.2
MCM (L=5)	58.3	66.5	54.4
kNN-ID‡	76.8	100.0	99.8
Mahalanobis-ID‡	82.4	100.0	100.0
PROTON	68.0	46.7	86.8
MCM (L=5) + PROTON	41.2	24.0	25.3
PROTON+ (ours)	67.3	85.3	93.4

Radiology

Method	Sem	Cov	Far
TAG-MCM	60.1	48.1	41.3
MCM (L=5)	56.1	78.4	84.6
kNN-ID‡	67.5	99.9	100.0
Mahalanobis-ID‡	60.3	100.0	100.0
PROTON	57.1	61.2	93.9
MCM (L=5) + PROTON	52.1	79.1	88.3
PROTON+ (ours)	65.2	93.9	91.5

Pathology

Method	Sem	Cov	Far
TAG-MCM	48.7	48.0	45.1
MCM (L=5)	15.3	64.6	61.1
kNN-ID‡	100.0	99.4	100.0
Mahalanobis-ID‡	100.0	99.8	100.0
PROTON	52.9	55.6	58.2
MCM (L=5) + PROTON	32.6	61.5	65.7
PROTON+ (ours)	72.5	70.1	81.1

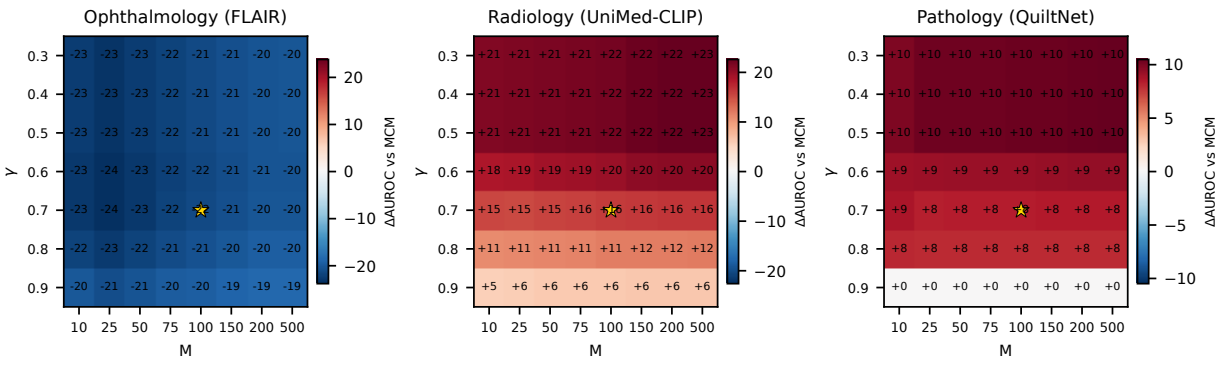


Figure 2: **A stricter gate helps PROTON only where ID images are more confident than the shifted ones.** Change in covariate AUROC of PROTON over MCM against γ and M, with a star at the defaults.

Table 16: Six base scores alone, with PROTON and with PROTON+ (AUROC Sem / Cov / Far), ophthalmology, radiology and pathology from top to bottom.

	Base			+PROTON			+PROTON+		
Base score	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
Max-Logits	54.0	27.9	83.9	57.3	23.6	87.0	57.1	87.1	99.1
Energy	51.8	27.2	83.9	55.3	22.6	87.0	56.1	87.2	99.1
GL-MCM	75.8	68.9	78.5	67.6	45.7	85.0	67.3	85.3	93.4
MCM	75.8	68.9	78.5	68.0	46.7	86.8	67.3	85.3	93.4
Entropy	75.8	68.9	78.5	70.0	49.7	89.8	63.3	86.5	96.6
TAG-MCM	68.4	30.1	72.2	67.9	22.6	89.4	65.2	58.7	91.5

	Base			+PROTON			+PROTON+		
Base score	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
Max-Logits	72.7	0.4	75.7	68.8	4.5	80.6	71.3	92.9	98.8
Energy	73.1	0.4	77.0	69.2	4.5	81.8	71.3	92.9	98.9
GL-MCM	56.8	46.7	29.2	56.9	60.6	93.5	65.2	93.9	91.6
MCM	56.8	46.6	27.6	57.1	61.2	93.9	65.2	93.9	91.5
Entropy	56.8	46.6	27.6	59.9	63.1	94.2	67.0	98.1	95.5
TAG-MCM	60.1	48.1	41.3	59.9	60.6	94.2	66.9	86.8	83.9

	Base			+PROTON			+PROTON+		
Base score	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
Max-Logits	7.1	24.5	17.7	11.3	29.3	24.6	75.7	81.5	93.7
Energy	6.7	23.9	17.5	10.9	28.6	24.5	75.6	81.9	93.6
GL-MCM	46.5	46.0	44.2	51.2	54.2	58.0	71.7	70.1	80.6
MCM	48.8	48.0	45.1	52.9	55.6	58.2	72.5	70.1	81.1
Entropy	48.8	48.0	45.1	55.2	57.6	61.9	74.8	72.4	82.8
TAG-MCM	48.7	48.0	45.1	52.9	55.6	58.2	72.5	70.1	81.1

Table 17: Bandwidth h of the domain anchor (AUROC).

	Ophthalmology			Radiology			Pathology		
h	Sem	Cov	Far	Sem	Cov	Far	Sem	Cov	Far
0.02	66.9	82.7	89.2	62.7	94.4	91.8	76.7	69.1	78.8
0.05	68.1	86.9	92.5	64.8	94.1	91.6	77.4	70.8	80.2
0.1	67.3	85.3	93.4	65.2	93.9	91.5	72.5	70.1	81.1
0.2	66.9	72.6	93.8	65.3	94.2	91.6	60.8	68.7	81.0
0.5	68.1	40.3	93.7	65.4	86.4	92.0	55.8	67.3	73.4

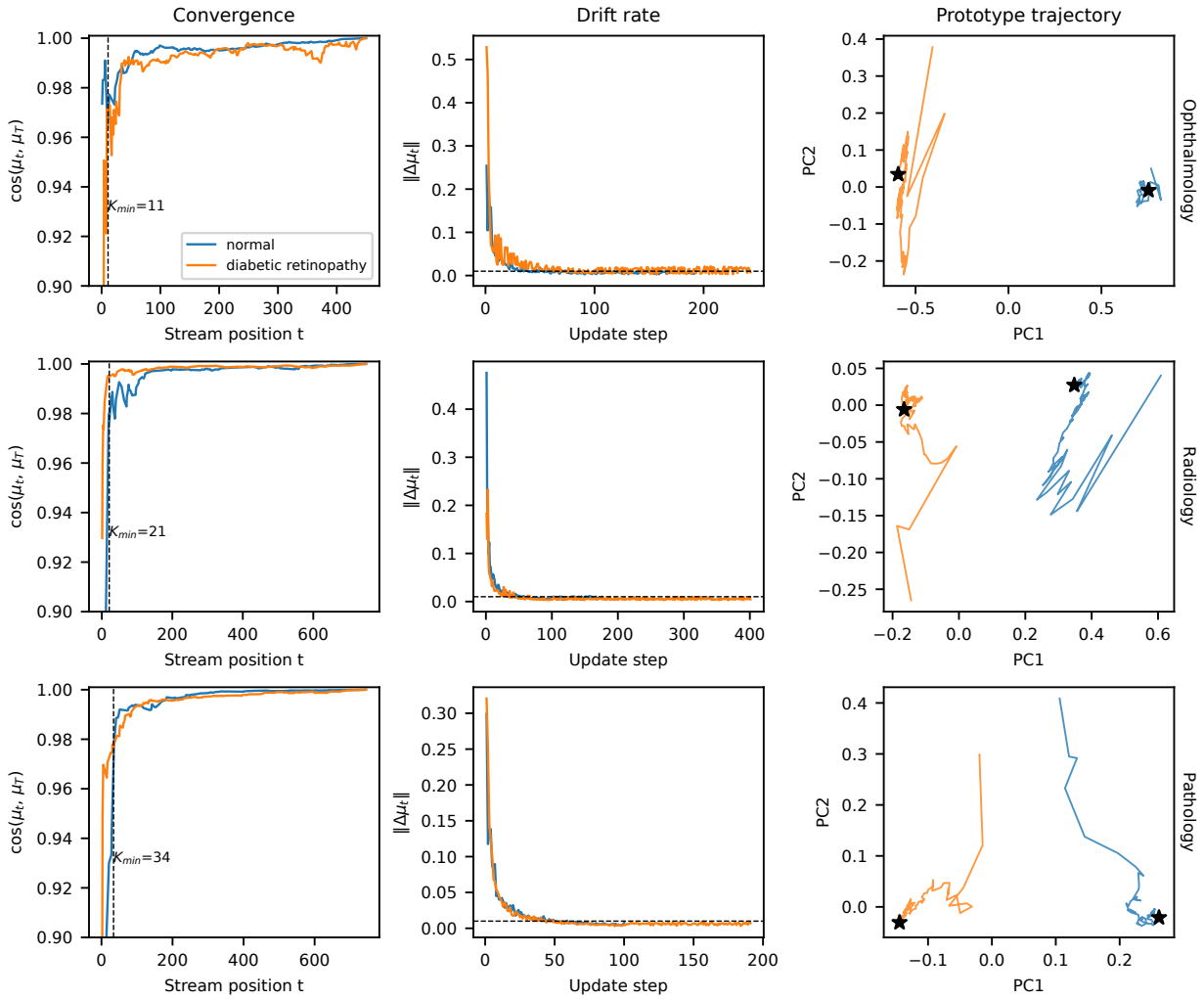


Figure 3: **Prototypes settle within the first hundred admissions.** Cosine similarity of the online prototype to its final value, drift per update, and PCA trajectories on the covariate stream, one row per modality.